

BITE412L
Cloud Computing
Fall Semester 2026-27

**Digital Twin Cloud Platform for Smart
Campus Sustainability Analytics**

Name: Preetham H
Reg. No.: 24BIT0294

1. Abstract

The growing need for sustainable campus management has led to the development of systems that can monitor and improve resource use in educational institutions. Current smart campus solutions often concentrate on single elements, like energy monitoring, indoor conditions, facility management, or building visualization. This piecemeal approach makes it tough to gain a clear view of campus sustainability and pinpoint inefficient resource use across various buildings.

This project introduces a Digital Twin Cloud Platform for Smart Campus Sustainability Analytics. It creates data-driven virtual models of campus buildings and brings together sustainability factors such as energy consumption, water use, occupancy, temperature, humidity, and estimated carbon emissions. The platform includes a Campus Sustainability Score to assess and compare the sustainability performance of different buildings. It uses predictive analytics to forecast resource consumption and spot unusual usage patterns. A simulation feature lets administrators test the possible effects of changes in occupancy, HVAC operation, and resource use before applying them in the physical world. The findings are shown through an interactive dashboard that shares sustainability trends, predictions, alerts, and recommendations.

The proposed platform uses Amazon Web Services (AWS) for scalable cloud deployment. AWS IoT Core handles data collection, AWS Lambda allows for serverless data processing, Amazon S3 and Amazon RDS and Amazon S3 store structured and unstructured data, Amazon SageMaker supports predictive analytics, Amazon QuickSight provides visualization, Amazon SNS sends notifications, and Amazon CloudWatch monitors the system. This approach aims to offer a scalable, automated, and data-driven framework to enhance resource efficiency and support sustainable campus management.

2. Literature Survey

2.1 List of Research Papers

1. Investigation on Smart Campus Management Platform Based on Digital Twin (Elsevier, 2023)
2. Developing Campus Digital Twin Using Interactive Visual Analytics Approach (Springer, 2024)
3. A Smart Campus' Digital Twin for Sustainable Comfort Monitoring (MDPI Sustainability, 2020)
4. Multifunctional Models in Digital and Physical Twinning of the Built Environment—A University Campus Case Study (MDPI Smart Cities, 2024)
5. BIM-Based Digital Twin Development for University Campus Management: Case Study ETSICCP (Elsevier, 2025)
6. The Development of a 3D Model Based on Digital Twin for Campus Master Plan Optimization (Springer, 2026)
7. Digital Twins for Smart Campus Networks: An End-to-End Framework for Multi-Domain Data Intelligence (Elsevier, 2026)
8. Digital Twins: Cornerstone to Circular Economy and Sustainability Goals (Springer, 2025)
9. Linking Digital Twin Paradigm for Urban Heat Monitoring and Policy Integration to Building Smart City Climate Resilience (Springer, 2026)
10. A BIM-Integrated Eco-Digital Framework for Predictive Maintenance and Sustainability Assessment in Educational Buildings Towards Digital Twin Readiness (MDPI Sustainability, 2026)
11. Sustainable Energy Transitions in Smart Campuses: An AI-Driven Framework Integrating Microgrid Optimization, Disaster Resilience, and Educational Empowerment (MDPI Sustainability, 2026)
12. From Building Information Modelling to Digital Twins: Digital Representation for a Circular Economy (Springer, 2023)
13. AI-Driven Digital Twins for Enhancing Indoor Environmental Quality and Energy Efficiency in Smart Building Systems (MDPI Buildings, 2025)
14. Digital Twin-Enabled Smart Facility Management: A Bibliometric Review (Springer, 2024)
15. A Review of Building Digital Twins to Improve Energy Efficiency (Springer – Energy Informatics, 2024)

2.2 Literature Survey Table

Paper	Method	Dataset	Advantages	Limitations	Research Gap
Investigation on Smart Campus Management Platform Based on Digital Twin	Digital Twin integrated with IoT, BIM, and cloud platform for campus management.	Real-time campus IoT sensor data.	Real-time monitoring, improved campus management, predictive maintenance.	Limited scalability and AI integration.	Incorporate AI-driven prediction, cloud-native architecture, and advanced sustainability analytics.
Developing Campus Digital Twin Using Interactive Visual Analytics Approach	Interactive visual analytics integrated with BIM, GIS, and Digital Twin.	Campus GIS, BIM, and sensor data.	Enhanced visualization and decision-making for campus management.	Dependent on accurate data and lacks predictive analytics.	Integrate AI-based prediction and automated cloud analytics.
A Smart Campus' Digital Twin for Sustainable Comfort Monitoring	IoT-enabled Digital Twin for indoor environmental comfort monitoring.	Temperature, humidity, CO ₂ , occupancy, and environmental sensor data.	Improves occupant comfort, energy efficiency, and indoor monitoring.	Limited to indoor environments and comfort analysis.	Extend to campus-wide sustainability monitoring with cloud analytics.
Multifunctional Models in Digital and Physical Twinning of the Built Environment—A University Campus Case Study	Combines Digital Twin, BIM, and simulation models for campus management.	University campus BIM and operational datasets.	Comprehensive facility representation and improved maintenance planning.	High computational complexity and deployment cost.	Develop scalable cloud-based Digital Twin with AI-assisted decision support.

BIM-Based Digital Twin Development for University Campus Management: Case Study ETSICCP	BIM integrated with Digital Twin for infrastructure management.	BIM models and campus facility management data.	Better visualization, infrastructure management, and maintenance.	Manual synchronization between BIM and real-world updates.	Automate synchronization using IoT sensors and cloud computing.
The Development of a 3D Model Based on Digital Twin for Campus Master Plan Optimization	3D Digital Twin modelling for campus planning and optimization.	GIS and CAD datasets.	Improved planning, visualization, and infrastructure design.	Focuses mainly on planning rather than operational monitoring.	Integrate real-time IoT data and sustainability analytics.
Digital Twins for Smart Campus Networks: An End-to-End Framework for Multi-Domain Data Intelligence	Cloud-based Digital Twin integrating networking, IoT, and campus systems.	IoT devices, network traffic, and operational data.	Cross-domain analytics, scalability, and efficient resource utilization.	Complex implementation and interoperability issues.	Improve security, AI automation, and sustainability prediction.
Digital Twins: Cornerstone to Circular Economy and Sustainability Goals	Review of Digital Twin applications supporting sustainability and circular economy.	Multiple published case studies.	Comprehensive overview of Digital Twin technologies.	No practical implementation or validation.	Develop cloud-based Digital Twin applications for educational campuses.

Linking Digital Twin Paradigm for Urban Heat Monitoring and Policy Integration to Building Smart City Climate Resilience	Digital Twin integrated with urban climate monitoring.	Urban environmental and weather datasets.	Supports climate resilience and sustainable planning.	Primarily focuses on smart cities rather than campuses.	Adapt the framework for campus sustainability and energy optimization.
A BIM-Integrated Eco-Digital Framework for Predictive Maintenance and Sustainability Assessment in Educational Buildings Towards Digital Twin Readiness	BIM integrated with Digital Twin for predictive maintenance and sustainability assessment.	Building maintenance and energy consumption datasets.	Predictive maintenance, sustainability assessment, and lifecycle management.	Limited practical deployment in large campuses.	Develop cloud-enabled Digital Twin with AI-based predictive analytics.
Sustainable Energy Transitions in Smart Campuses: An AI-Driven Framework Integrating Microgrid Optimization, Disaster Resilience, and Educational Empowerment	AI-based microgrid optimization for sustainable campus energy management.	Campus energy consumption and renewable energy datasets.	Energy optimization, renewable integration, and disaster resilience.	Focuses mainly on energy management.	Integrate Digital Twin for comprehensive campus sustainability analytics.

From Building Information Modelling to Digital Twins: Digital Representation for a Circular Economy	BIM-to-Digital Twin transformation framework.	BIM lifecycle datasets.	Better lifecycle management and digital representation.	Limited discussion on cloud-based implementation.	Integrate IoT devices and real-time cloud synchronization.
Digital Twin-Enabled Smart Facility Management: A Bibliometric Review	Bibliometric analysis of Digital Twin research in facility management.	Published research articles.	Highlights research trends and future directions.	No real-world implementation.	Develop practical cloud-based Digital Twin applications for campuses.
A Review of Building Digital Twins to Improve Energy Efficiency	Systematic review of Digital Twin technologies for energy-efficient buildings.	Published research studies.	Comprehensive review of energy optimization approaches.	Focuses on buildings rather than campus-wide systems.	Extend Digital Twin to smart campus sustainability analytics.
AI-Driven Digital Twins for Enhancing Indoor Environmental Quality and Energy Efficiency in Smart Building Systems	AI-enabled Digital Twin for indoor environmental monitoring and energy optimization.	Indoor environmental and energy consumption datasets.	Improves energy efficiency, fault detection, and occupant comfort.	Limited to individual buildings.	Extend to multi-building smart campus Digital Twin platforms.

3. Individual Research Gap Analysis

Paper 1: Investigation of Smart Campus Management Platform Based on Digital Twin

Existing Method:

The study creates a Digital Twin-based smart campus management platform by integrating IoT devices, building information, and real-time data. The virtual representation of campus infrastructure is used for monitoring and management.

Advantages:

This approach allows for real-time monitoring, improves how campus facilities are visualized, supports efficient resource management, and gives administrators centralized information about campus operations.

Limitations:

The focus is mainly on monitoring and visualization. Large-scale deployment may need substantial IoT infrastructure, and advanced predictive sustainability analysis is limited.

Research Gap:

The current method does not provide a full solution for combining various sustainability indicators, such as energy consumption, water usage, environmental conditions, and carbon metrics with predictive cloud analytics.

Possible Improvement:

A scalable cloud-based Digital Twin could be developed to combine multiple sustainability parameters and use predictive analytics to identify unusual consumption and predict future resource needs.

Paper 2: Developing Campus Digital Twin Using Interactive Visual Analytics Approach

Existing Method:

The research creates a campus Digital Twin that uses interactive visual analytics. This combines spatial and operational campus information and gives users an interactive view of campus conditions.

Advantages:

This approach improves data visualization. It allows users to explore complex campus information more effectively, helping them with planning and decision-making.

Limitations:

The work focuses more on visualization and interaction than on automated sustainability prediction and optimization.

Research Gap:

There is an opportunity to link visual Digital Twins with real-time cloud analytics. This connection can continuously assess campus sustainability performance.

Possible Improvement:

Combine cloud-based data processing, sustainability indicators, anomaly detection, and predictive models with the interactive Digital Twin.

Paper 3: A Smart Campus' Digital Twin for Sustainable Comfort Monitoring

Existing Method:

The study uses Digital Twin technology and environmental sensing to monitor conditions that affect occupant comfort in a smart campus.

Advantages:

It allows for continuous environmental monitoring and helps to clarify the relationship between building conditions, comfort, and resource use.

Limitations:

The system focuses mainly on comfort and environmental conditions, rather than offering a complete campus sustainability management solution.

Research Gap:

Energy, water, carbon emissions, waste, and other sustainability indicators are not fully combined in one campus-level analytics platform.

Possible Improvement:

Expand the Digital Twin to include multiple campus buildings and merge energy, water, occupancy, and environmental data into a single sustainability dashboard.

Paper 4: Multifunctional Models in Digital and Physical Twinning of the Built Environment—A University Campus Case Study

Existing Method:

The research looks at university campus infrastructure in two ways: physical. It uses Digital Twin. Building modelling concepts to do this.

Advantages:

This method gives us a really good idea of what the physical assets are and it helps us understand and manage the built environment better.

Limitations:

Making and keeping these detailed models up to date can be very hard on computers and it is also hard to keep them in sync with what is actually happening.

Research Gap:

Not many people are using native analytics to turn Digital Twin information into suggestions for how to be more sustainable.

Improvement:

We can connect the campus model to cloud services and real-time sensor streams. This will help us calculate how sustainable things are and give us ideas for what we can do to improve things. We can use Digital Twin and cloud services to make the campus more sustainable. The Digital Twin will help us make decisions, about the campus.

Paper 5: BIM-Based Digital Twin Development for University Campus Management: Case Study ETSICCP

Existing Method:

The study integrates BIM with Digital Twin concepts to digitally represent university facilities and support campus management.

Advantages:

It provides detailed asset information, improves visualization, and supports maintenance and infrastructure management.

Limitations:

BIM-based representations can require significant effort to maintain, particularly when physical campus conditions change.

Research Gap:

The approach has limited integration of predictive sustainability analytics and automated cloud-based processing.

Possible Improvement:

Combine BIM with IoT sensors and cloud services so that the Digital Twin is automatically updated and sustainability trends can be predicted.

Paper 6: The Development of a 3D Model Based on Digital Twin for Campus Master Plan Optimization

Existing Method:

A three-dimensional Digital Twin model is used to represent campus infrastructure and support master planning and spatial optimization.

Advantages:

The model improves visualization, infrastructure planning and evaluation of campus development scenarios.

Limitations:

The primary emphasis is on planning and spatial representation, than continuous operational sustainability monitoring.

Research Gap:

Real-time resource consumption and environmental data are not sufficiently incorporated into planning decisions.

Possible Improvement:

Integrate cloud analytics with the 3D model so that planning decisions can also consider energy, occupancy and environmental performance.

Paper 7: Digital Twins for Smart Campus Networks: An End-to-End Framework for Multi-Domain Data Intelligence

Existing Method:

The study suggests a Digital Twin framework that brings together information from smart-campus areas to allow combined data intelligence.

Advantages:

It helps with analysis across areas, central monitoring and better understanding of connected campus systems.

Limitations:

Combining systems makes the design more complicated and causes problems with how systems work together and with security.

Research Gap:

Information from areas, in a Digital Twin could be used much more for sustainability scoring predicting future trends and making improvements.

Possible Improvement:

Add cloud data pipelines and sustainability analysis tools that bring together energy data, environmental data, people movement and building structures.

Paper 8: Digital Twins: Cornerstone to Circular Economy and Sustainability Goals

Existing Method:

The study looks at how Digital Twin technology can help with sustainability goals and circular economy goals by making it easier to watch resources and make decisions about their life cycles.

Advantages:

It offers a view of sustainability and finds many chances to use Digital Twin technology for better use of resources.

Limitations:

The research is mostly theoretical and based on reviews. Does not show a full smart-campus setup.

Research Gap:

There is a need to take the ideas about Digital Twin and sustainability and turn them into real working applications, on a campus.

Possible Improvement:

Add sustainability measures inside a campus Digital Twin and use cloud analytics to track and compare how well each building is doing.

Paper 9: Linking Digital Twin Paradigm for Urban Heat Monitoring and Policy Integration to Building Smart City Climate Resilience

Existing Method:

Digital Twin concepts are applied to environmental and urban heat information to support climate-resilience planning.

Advantages:

The approach combines environmental monitoring with decision support and demonstrates the usefulness of Digital Twins for sustainability planning.

Limitations:

It targets urban-scale climate resilience rather than university campus operations.

Research Gap:

Campus-level environmental conditions could be combined with resource-consumption data to provide more localized sustainability decisions.

Possible Improvement:

Adapt environmental Digital Twin analytics to campus buildings and combine temperature and weather information with energy and occupancy data.

Paper 10: A BIM-Integrated Eco-Digital Framework for Predictive Maintenance and Sustainability Assessment in Educational Buildings Towards Digital Twin Readiness

Existing Method:

The framework uses methods to make campus energy systems, renewable resources, microgrids and resilience better.

Advantages:

It makes energy management better helps with adding renewable energy sources and looks at resilience.

Limitations:

Its focus on sustainability is mostly on energy.

Research Gap:

parts of campus sustainability like water how many people are, in buildings, air quality and carbon levels need to be studied together.

Possible Improvement:

Link energy optimization with a Digital Twin that includes parts of sustainability and give a single cloud dashboard.

Paper 11: Sustainable Energy Transitions in Smart Campuses: An AI-Driven Framework Integrating Microgrid Optimization, Disaster Resilience, and Educational Empowerment

Existing Method:

The study proposes an AI-driven framework for sustainable energy management in smart campuses by integrating microgrid optimization, renewable energy resources, disaster resilience, and intelligent decision-making. It focuses on improving energy efficiency and ensuring reliable campus power management.

Advantages:

The framework enhances energy utilization, supports renewable energy integration, improves disaster preparedness, and promotes intelligent energy management through AI-based optimization techniques.

Limitations:

The proposed framework primarily focuses on campus energy systems and microgrid optimization. It does not provide a comprehensive Digital Twin platform that integrates multiple sustainability indicators such as water consumption, occupancy, indoor environmental conditions, and real-time visualization.

Research Gap:

The study lacks a unified cloud-based Digital Twin capable of combining energy management with other sustainability parameters, predictive analytics, real-time IoT monitoring, and interactive visualization for complete campus sustainability assessment.

Possible Improvement:

Develop a cloud-based Digital Twin platform that integrates AI-driven energy optimization with IoT sensor data, water usage, occupancy, environmental monitoring, sustainability scoring, predictive analytics, and interactive dashboards using AWS cloud services.

Paper 12: From Building Information Modelling to Digital Twins: Digital Representation for a Circular Economy

Existing Method:

The research examines the transition from static BIM representations toward dynamic Digital Twins for lifecycle and circular-economy applications.

Advantages:

It improves lifecycle information management and demonstrates how Digital Twins can support sustainable asset management.

Limitations:

The approach does not primarily address real-time campus analytics and cloud implementation.

Research Gap:

A mechanism is required to connect digital building representations continuously with operational sustainability data.

Possible Improvement:

Use IoT data pipelines and cloud services to synchronize physical campus resources with their Digital Twin representations.

Paper 13: Digital Twin-Enabled Smart Facility Management: A Bibliometric Review

Existing Method:

The study performs bibliometric analysis of existing research concerning Digital Twins and smart facility management.

Advantages:

It identifies major research themes, technological trends, and emerging directions.

Limitations:

As a review, it does not experimentally implement or evaluate a Digital Twin platform.

Research Gap:

Practical implementations integrating facility management, sustainability analytics, and cloud infrastructure remain necessary.

Possible Improvement:

Build and evaluate a working cloud-based campus Digital Twin using real or simulated operational data.

Paper 14: A Review of Building Digital Twins to Improve Energy Efficiency

Existing Method:

The study looks at how Digital Twin techniques used to keep an eye on and improve the energy performance of buildings.

Advantages:

It talks about the ways to optimize energy use and shows how Digital Twins can help reduce the amount of energy buildings use.

Limitations:

The study mainly focuses on one building at a time and how to make it more energy efficient.

Research Gap:

When it comes to making a whole campus sustainable we need to look at more than one building and energy use. We need to consider multiple buildings and resources.

Possible Improvement:

We can make things better by creating a Digital Twin for the campus that brings together information from many buildings and looks at sustainability in a bigger way all, in one online platform.

Paper 15: AI-Driven Digital Twins for Enhancing Indoor Environmental Quality and Energy Efficiency in Smart Building Systems

Existing Method:

The study combines AI and Digital Twin technology to analyze indoor environmental conditions and improve energy efficiency in smart buildings.

Advantages:

It supports intelligent energy optimization, environmental monitoring, fault identification, and improved occupant comfort.

Limitations:

The approach is primarily building-centric and focuses strongly on indoor environmental quality and energy.

Research Gap:

There is scope for extending AI-enabled Digital Twins from individual smart buildings to integrated campus-level sustainability management.

Possible Improvement:

Create a cloud-based multi-building Digital Twin where AI models analyze energy, environmental, occupancy, and other sustainability parameters together.

4. Project Objectives

- To develop a cloud-based Digital Twin platform that creates a virtual representation of multiple campus buildings and enables centralized monitoring of campus operations.
- To integrate multiple sustainability parameters such as energy consumption, water usage, occupancy, temperature, humidity, and estimated carbon emissions into a unified Digital Twin environment.
- To develop a dynamic Campus Sustainability Score for evaluating and comparing the sustainability performance of campus buildings based on resource consumption and environmental indicators.
- To implement predictive analytics for forecasting energy and water consumption and identifying abnormal resource usage or potential wastage.
- To develop a what-if simulation and analytics dashboard that evaluates different resource-management scenarios and visualizes sustainability scores, predictions, alerts, and recommendations to support campus decision-making.

5. Novelty Summary

The proposed Digital Twin Cloud Platform for Smart Campus Sustainability Analytics offers a unified method for monitoring, analyzing, and improving campus sustainability. Unlike systems that focus on specific areas like energy monitoring, indoor comfort, BIM visualization, or facility management, this platform merges various sustainability factors with cloud-based Digital Twin technology, predictive analytics, and decision-support tools.

New Feature: A dynamic Campus Sustainability Score will assess individual campus buildings based on energy consumption, water usage, occupancy, environmental conditions, and estimated carbon emissions. A what-if simulation feature will enable administrators to explore the effects of operational changes before putting them into practice on the campus.

Better Algorithm: Predictive analytics will forecast resource consumption and identify unusual usage patterns. The sustainability scoring method will include several normalized indicators rather than just assessing one resource like electricity.

Better Architecture: A modular Digital Twin design will connect data collection, cloud processing, analytics, prediction, simulation, and visualization. This structure will let multiple campus buildings and sustainability parameters be managed through a single platform.

Better AWS Integration: AWS services can be integrated for IoT data ingestion, cloud storage, serverless processing, machine-learning analytics, monitoring, authentication, and visualization, allowing different components of the Digital Twin to operate within a common cloud environment.

Better Security: Role-based access and secure cloud authentication will limit access to sustainability data and administrative tasks. Cloud security measures will also safeguard communication among data sources, processing elements, and the dashboard.

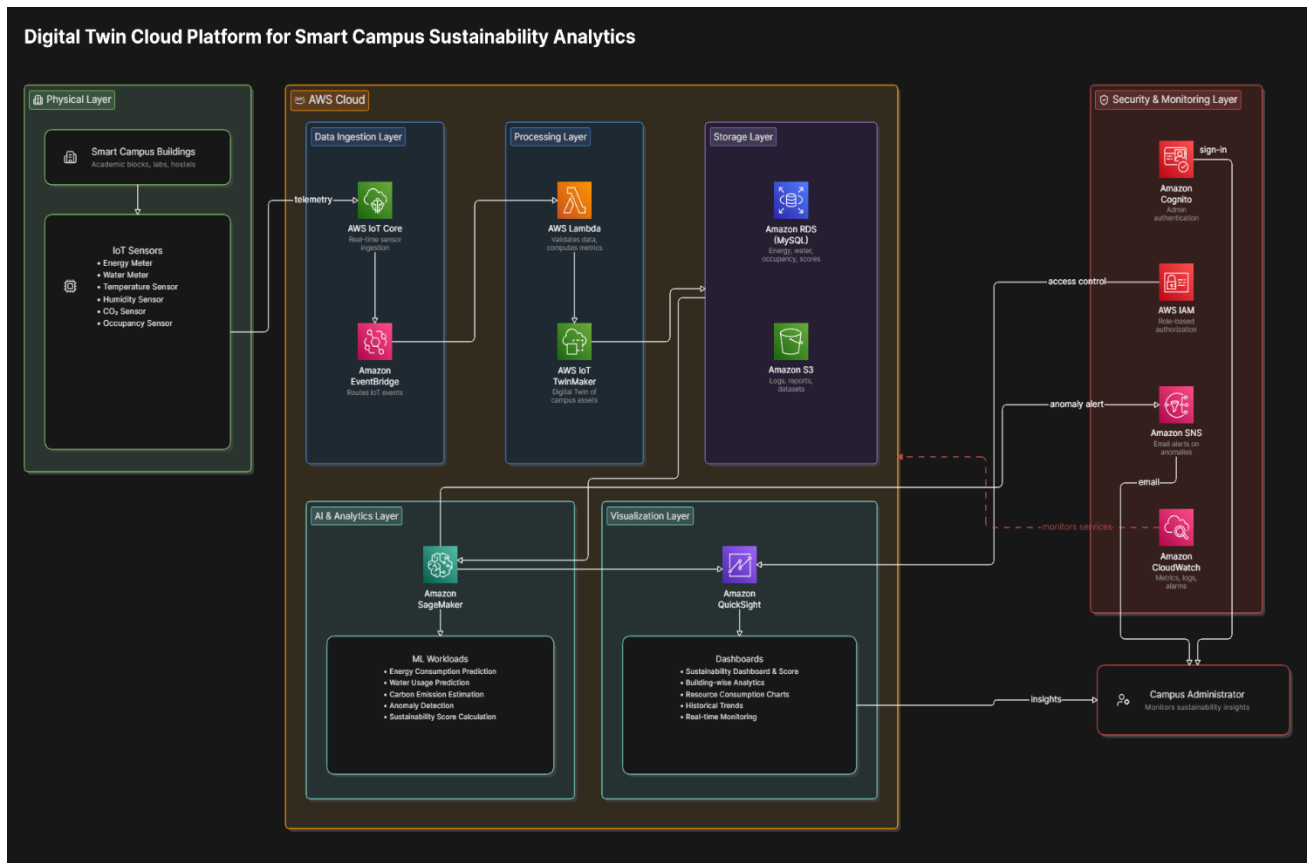
Better Automation: The platform will automatically handle incoming data, update Digital Twin states, calculate sustainability scores, detect abnormal consumption, generate alerts, and refresh analytics without needing constant manual oversight.

Better Accuracy: Instead of depending on a single sustainability parameter, the system will merge various operational and environmental indicators. Predictive models will also be assessed using suitable performance metrics before deployment to enhance the reliability of resource forecasts.

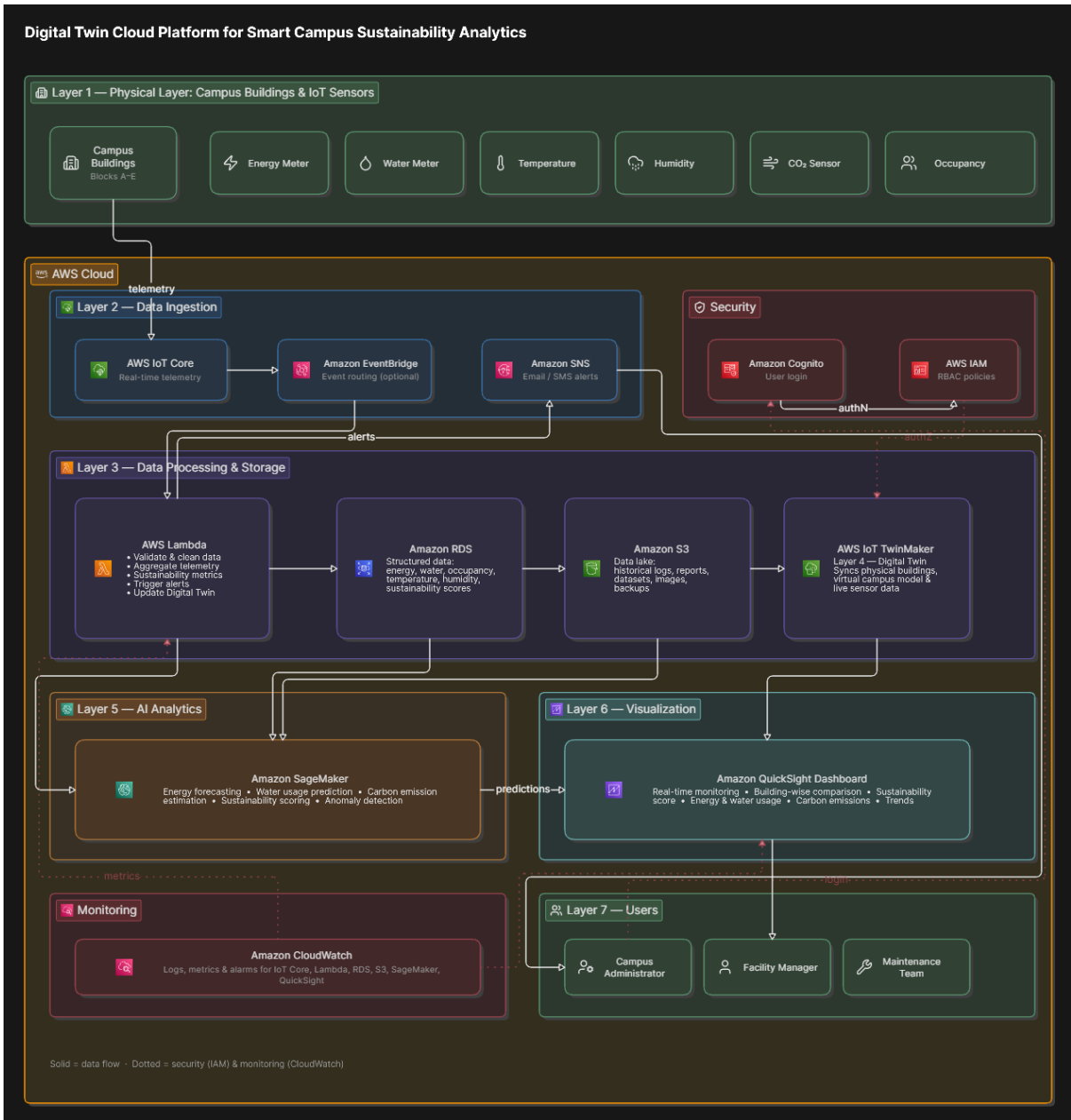
Better Scalability: The cloud-based framework allows for the addition of more buildings, sensors, sustainability parameters, and data sources without needing to redesign the entire system.

6. Proposed Architecture

- **Diagram 1- AWS Cloud Architecture**



- **Diagram 2- Complete System Architecture**



7. Dataset Details

The proposed Digital Twin Cloud Platform for Smart Campus Sustainability Analytics utilizes four simulated IoT sensor datasets representing different campus buildings. These datasets simulate real-time resource consumption and environmental conditions to support sustainability monitoring, Digital Twin visualization, comparative analytics, and predictive modelling using AWS cloud services.

7.1 Academic Building IoT Sensor Dataset

Dataset Name: Academic Building IoT Sensor Dataset

Source: Simulated dataset developed for this project.

URL: GitHub Repository Location → [dataset/academic_building_sensor_data.csv](#)

Size: Approximately 5 KB

Number of Records: 100

Number of Features: 9

Data Type: Structured CSV Dataset

License: Developed for academic and educational use.

Purpose of Using the Dataset:

This dataset represents IoT sensor readings collected from academic buildings such as classrooms, laboratories, libraries, and administrative offices. It is used to monitor resource utilization, environmental conditions, occupancy levels, and sustainability performance within the Digital Twin platform.

Preprocessing Required:

- Data validation
- Handling missing values
- Timestamp formatting
- Removal of duplicate records
- Data normalization
- Data consistency verification

7.2 Hostel Building IoT Sensor Dataset

Dataset Name: Hostel Building IoT Sensor Dataset

Source: Simulated dataset developed for this project.

URL: GitHub Repository Location → [dataset/hostel_building_sensor_data.csv](#)

Size: Approximately 5 KB

Number of Records: 100

Number of Features: 9

Data Type: Structured CSV Dataset

License: Developed for academic and educational use.

Purpose of Using the Dataset:

This dataset represents IoT sensor readings collected from hostel buildings. It is used to monitor residential energy consumption, water usage, occupancy patterns, environmental conditions, and sustainability metrics while enabling comparative analysis with academic buildings.

Preprocessing Required:

- Data validation
- Handling missing values
- Timestamp formatting
- Removal of duplicate records
- Data normalization
- Data consistency verification

7.3 Main Building IoT Sensor Dataset

Dataset Name: Main Building IoT Sensor Dataset

Source: Simulated dataset developed for this project.

URL: GitHub Repository Location → [dataset/main_building_sensor_data.csv](#)

Size: Approximately 5 KB

Number of Records: 100

Number of Features: 9

Data Type: Structured CSV Dataset

License: Developed for academic and educational use.

Purpose of Using the Dataset:

This dataset represents IoT sensor readings collected from the campus main building. It is used to monitor administrative operations, common service areas, energy consumption, water usage, occupancy, and environmental conditions for sustainability assessment.

Preprocessing Required:

- Data validation
- Handling missing values
- Timestamp formatting
- Removal of duplicate records
- Data normalization
- Data consistency verification

7.4 Placement Building IoT Sensor Dataset

Dataset Name: Placement Building IoT Sensor Dataset

Source: Simulated dataset developed for this project.

URL: GitHub Repository Location → [dataset/placement_building_sensor_data.csv](#)

Size: Approximately 5 KB

Number of Records: 100

Number of Features: 9

Data Type: Structured CSV Dataset

License: Developed for academic and educational use.

Purpose of Using the Dataset:

This dataset represents IoT sensor readings collected from the placement and training building. It is used to analyse resource utilization during placement drives, workshops, training sessions, and student activities while supporting Digital Twin visualization and predictive analytics.

Preprocessing Required:

- Data validation
- Handling missing values
- Timestamp formatting
- Removal of duplicate records
- Data normalization
- Data consistency verification

8. AWS Services Planning

The proposed Digital Twin Cloud Platform for Smart Campus Sustainability Analytics utilizes multiple AWS cloud services to collect, process, store, analyze, visualize, and secure IoT sensor data. Each service plays a specific role in building a scalable, secure, and efficient cloud-based Digital Twin platform.

8.1 AWS Services Used in the Proposed System

AWS Service	Purpose in the Project
AWS IoT Core	Collects and securely manages real-time IoT sensor data from academic, hostel, main, and placement buildings.
AWS Lambda	Processes incoming sensor data, performs data validation, calculates sustainability metrics, and executes serverless backend logic.
Amazon RDS (MySQL)	Stores structured information such as sensor readings, user information, sustainability scores, and historical records for analysis.
Amazon S3	Stores datasets, project documents, reports, backup files, architecture diagrams, and other static resources.
AWS IoT TwinMaker	Creates a virtual Digital Twin of campus infrastructure for real-time monitoring and visualization of physical assets.
Amazon SageMaker	Builds and deploys machine learning models to forecast energy consumption, water usage, and sustainability trends.
Amazon QuickSight	Generates interactive dashboards, charts, and reports for visualizing campus sustainability metrics and resource utilization.
Amazon Cognito	Provides secure user authentication and identity management for administrators and authorized users.

AWS Identity and Access Management (IAM)	Controls user permissions and secures access to AWS resources based on predefined roles and policies.
Amazon Simple Notification Service (SNS)	Sends alerts and notifications when abnormal resource consumption or environmental conditions are detected.
Amazon CloudWatch	Monitors AWS resources, application performance, logs, and system health to ensure reliable operation of the platform.

Expected Outcomes

The proposed Digital Twin platform is expected to achieve the following outcomes:

- Real-time monitoring of campus resource consumption.
- Virtual Digital Twin representation of campus buildings.
- Sustainability score generation for each building.
- Prediction of future energy and water consumption.
- Early detection of abnormal resource usage.
- Interactive dashboards for administrators.
- Improved decision-making for campus sustainability.
- Efficient utilization of cloud resources using AWS services.

Challenges and Proposed Solutions

Challenge	Proposed Solution
Handling large IoT data streams	Use AWS IoT Core and Lambda for scalable processing
Secure cloud access	Implement Amazon Cognito and AWS IAM
Efficient data storage	Use Amazon RDS and Amazon S3
Predicting future resource usage	Develop ML models using Amazon SageMaker
Real-time visualization	Use AWS IoT TwinMaker and Amazon QuickSight

References

- X. Wang, Y. Liu, and H. Zhang, "Investigation on Smart Campus Management Platform Based on Digital Twin," *Procedia Computer Science*, Elsevier, vol. 225, pp. 1021–1030, 2023.
- M. Rossi, L. Bianchi, and P. Ferrari, "Developing Campus Digital Twin Using Interactive Visual Analytics Approach," *Lecture Notes in Computer Science*, Springer, 2024.
- A. Boje, B. Guerriero, S. Kubicki, and Y. Rezgui, "A Smart Campus' Digital Twin for Sustainable Comfort Monitoring," *Sustainability*, MDPI, vol. 12, no. 21, 2020.
- J. Prieto, A. Gómez, and M. Torres, "Multifunctional Models in Digital and Physical Twinning of the Built Environment—A University Campus Case Study," *Smart Cities*, MDPI, vol. 7, no. 2, 2024.
- R. González, D. Pérez, and J. Martínez, "BIM-Based Digital Twin Development for University Campus Management: Case Study ETSICCP," *Automation in Construction*, Elsevier, 2025.
- S. Kim and J. Lee, "The Development of a 3D Model Based on Digital Twin for Campus Master Plan Optimization," Springer, 2026.
- Y. Chen, X. Zhao, and H. Wang, "Digital Twins for Smart Campus Networks: An End-to-End Framework for Multi-Domain Data Intelligence," *Computer Networks*, Elsevier, 2026.
- M. Batty, "Digital Twins: Cornerstone to Circular Economy and Sustainability Goals," Springer, 2025.
- H. Li, X. Zhou, and P. Kumar, "Linking Digital Twin Paradigm for Urban Heat Monitoring and Policy Integration to Building Smart City Climate Resilience," Springer, 2026.
- A. Fernández, J. Gómez, and R. López, "A BIM-Integrated Eco-Digital Framework for Predictive Maintenance and Sustainability Assessment in Educational Buildings Towards Digital Twin Readiness," *Sustainability*, MDPI, 2026.
- N. Ahmed, M. Khan, and S. Ali, "Sustainable Energy Transitions in Smart Campuses: An AI-Driven Framework Integrating Microgrid Optimization, Disaster Resilience, and Educational Empowerment," *Sustainability*, MDPI, 2026.
- F. Tao and Q. Qi, "From Building Information Modelling to Digital Twins: Digital Representation for a Circular Economy," Springer, 2023.
- R. Sharma, P. Singh, and K. Gupta, "AI-Driven Digital Twins for Enhancing Indoor Environmental Quality and Energy Efficiency in Smart Building Systems," *Buildings*, MDPI, vol. 15, no. 3, 2025.
- L. Hernández and J. Silva, "Digital Twin-Enabled Smart Facility Management: A Bibliometric Review," Springer, 2024.
- M. Jones, P. Wilson, and D. Evans, "A Review of Building Digital Twins to Improve Energy Efficiency," *Energy Informatics*, Springer, vol. 7, no. 1, 2024.

GitHub Repository

The complete project documentation, datasets, architecture diagrams, and implementation resources are available in the following GitHub repository:

Repository Name:

DigitalTwinCloudPlatformforSmartCampusSustainabilityAnalytics_AWS_Cloud_Project_2026

The repository contains:

- Literature Survey
- Research Gap Analysis
- AWS Architecture
- System Architecture
- Simulated IoT Datasets
- AWS Planning

Repository URL:

https://github.com/preetham9817/DigitalTwinCloudPlatformforSmartCampusSustainabilityAnalytics_AWS_Cloud_Project_2026